Program 4

Project Structure:

lab4
│
├── app.py
├── requirements.txt
├── Dockerfile
└── README.md

Dockerfile:

```
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

app.py

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
   return "hello world from container"

if __name__ == '__main__':
   app.run(host="0.0.0.0", port=5000)
```

Requirements.txt

Flask==2.3.3

Setup and Installation:

Docker build command:

```
$ sudo docker build -t program4 .
[+] Building 0.7s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 199B
=> [internal] load metadata for docker.io/library/python:lates
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 260B
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:9726101fc3e2556a3e79677ba73114e1e82
```

Run the docker on port:

```
$ sudo docker run -p 5000:5000 program4
* Serving Flask app 'app'
* Debug mode: off
ARNING: This is a development server. Do not use it in a production deployme
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
ress CTRL+C to quit
72.17.0.1 - - [07/Nov/2025 04:19:53] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [07/Nov/2025 04:19:56] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [07/Nov/2025 04:23:05] "GET / HTTP/1.1" 200 -
72.17.0.1 - - [07/Nov/2025 04:23:05] "GET /favicon.ico HTTP/1.1" 404 -
72.17.0.1 - - [07/Nov/2025 04:26:31] "GET / HTTP/1.1" 200 -
C
```

Open:
http://localhost:5000

hello world

Now, log in to you github and generate pat:

How to Generate a Personal Access Token (Classic)

Log in to your GitHub account.

First create a new repo then,
Go to Settings → Developer settings → Personal access tokens → Tokens (classic)
or visit directly:
https://github.com/settings/tokens

Click Generate new token → Generate new token (classic)

Enter a note (e.g., *Git CLI Access*) and select expiration time.

Under Select scopes, check:

- **`repo`→ (Full control of private repositories)**
- **`workflow` → (If you use GitHub Actions)**

Click Generate token

Step 1: Initialize a Local Repository

Create or move into your project directory:

cd ~/lab4demo

Initialize a new Git repository:

git init

Rename the default branch to `main`:

git branch -M main

Step 2: Add and Commit Files

git add .

Commit them with a message:

git commit -m "Initial commit"

Step 3: Connect to a Remote GitHub Repository

git remote add origin https://github.com/<your-username>/<repo-name>.git

Step 4. Using Username and PAT with Git

git push -u origin main

You will be prompted:

Username for 'https://github.com': <your username of github>
Password for 'https://kuntiwari1@github.com': <paste your PAT>